Video Transcript

Video 8: Introduction to Web Sockets (04:29)

Let's talk about web sockets. We've been using HTTP which runs over TCP networks, but web sockets also run over TCP networks. Now, web sockets provide a persistent connection, HTTP doesn't. That HTTP is a request response protocol. TCP sets up a persistent two-way connection, and either or both parties can then send messages. Now, it requires both parties to set up a TCP connection, but either of them can then take it down. It's heavily used where we have streaming data, for example, in IoT where we have sensors. Here, we're showing a number of sensors. For example, perhaps a drone is sending video feed, and that a number of clients may be interested in receiving data from those feeds.

So, web sockets is often used in chat messaging apps. It's a lighter protocol than HTTP, so it doesn't have so much in the headers. And it's preferred for, say, IoT networks where we're sending a lot of streaming messages. And we've seen how to handle large files of over a hundred million lines. But, how about streams of data that continue for weeks, months, or years? How do we handle those streams of data? Well, web sockets can help us. It's not the only one. We're going to look at another one called MQTT, but web sockets is widely used.

So, just to remind you about TCP networks. So, TCP packets have the address that they're going to be sent to, and it's an IP address plus a port number. And it has an IP address of where it's being sent from and also the port number there. Now, a machine has many processes listening on ports, and some of those ports may be listening for TCP messages, but there are also other message types, UDP, FTP, DNS, etc. So, we have these apps continuously listening on these ports, and here's some of the other protocols that you might see, tcp, tcp6, udp.

If you do netstat in a terminal window, you'll see who your machine is talking to, and you'll be surprised the number of communications it's got set up. Now, the TCP messages are coming in a stream, and they're directed to certain ports. Now, if it's a TCP port, then it will have a buffer. And the packets are not in order as they arrive, but the facilities on your machine will order those TCP packets in the TCP buffers. Now this is going to cause us some issues because the web socket

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protocol doesn't have the concept of a message.

So, the stream of packets are going to be coming in, they'll fill up the TCP buffer and when we make a web socket request, a read for example, it'll read the whole buffer. So, we're going to have to handcraft into the web services protocol, the concept of a message, so that we know where messages begin and where they end. We need to know when are we done with this message, and when do we need to wait, and then read another message. So, let's take a look at setting up a web socket messaging system where we have persistent connections, and we have streams of data that may be running for days, or weeks, or months. Okay, let's go and code messages into web sockets. Bye for now.

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